Technical Evaluation Report

Release 2026.10.v2 | Synthetic data | Seed 20261014

Release and reproducibility

Application release 2026.10.v2; dataset version 2026.10.v1; synthetic generator seed 20261014; 8 branches, 60 technicians, 3,451 orders. Automated tests verified in the current CI run: 49 passing. The exact CI log is published under juniper-command-center/docs/current-tests.txt.

Calculation boundaries

Branch-level financial statements reconcile by construction to consolidated totals. Each branch applies demand, available labor minutes, average service duration, travel time and data eligibility before projecting completed services. Revenue and direct costs derive from projected completed counts and modeled unit economics. Branch fixed costs are deducted once.

Mathematical optimization

The bounded OR-Tools CP-SAT solver evaluated 12 first-day orders with 8 branch technicians. Status: OPTIMAL. It assigned 11 work orders compared with 11 assigned by the reproducible greedy baseline; elapsed solver time 2.9544 seconds. The model enforces technician skills, windows, shift limits, inventory and time between jobs. It uses a synthetic neighborhood travel proxy rather than live road-network time; no full-fleet VRPTW benchmark is claimed.

Program scheduling and quantitative risk

The program scheduler uses day-level dependency precedence and shared technical resource slots. The seeded Monte Carlo model has 400 draws, with P10/P50/P90 finish days 52/57/62. Inputs are independently sampled triangular task-duration and cost multipliers; real dependencies between risk drivers are not estimated.

Investment and benefits

Four initiatives: CRM, DISPATCH, AI, ADOPTION. CRM and adoption are mandatory; available funds and resource capacity constrain eligible combinations. Under the complete $202,000 budget, recommended portfolio NPV is $284,257. Discounts, adoption assumptions and overlap adjustments are source-controlled. NPV is not booked savings.

Workflow transaction integrity

Synthetic appointments and incident proposals persist to SQLite. Mutations require a server-verified bearer token and an approved state transition. Execution checks CRM quality, incident completeness, an eligible target record and the recorded appointment version. Updates and audit events commit together; stale proposals are rejected. The public site has no anonymous privileged writes.

AI interpretation and evaluation

Optional OpenAI chat-completion integration validates structured JSON against a required schema and forces human review for ambiguous or injected instructions. Credentials remain server-side. The 35-case reference evaluation (version 2026.10.v2-35) has NOT been run against a live provider in this release because no authorized API credential or usage budget was supplied. A distinct 20-case deterministic replay set is not presented as LLM accuracy.

Deployment and known limitations

Backend is locally runnable via FastAPI, not currently deployed to a public host. The public calculator implements aggregate scenario math; it does not host OR-Tools or execute CRM transactions. Token authentication is suitable for a controlled local demonstration but not a full identity platform. Synthetic data, independent risk draws, limited route size and generic travel proxies restrict external validity.

Reproduction

From juniper-command-center: pip install -r requirements.txt; python -m lab.web_export; pytest -q; uvicorn lab.api:app --reload. Privileged actions require JUNIPER_MANAGER_TOKEN or JUNIPER_DISPATCHER_TOKEN configured as a sufficiently long secret.